IRRIGATION WATER MANAGEMENT · NOTE SP04

Crop Water and Irrigation Requirement — Water Requirement, Duty and Delta, and Irrigation Scheduling

Course topics: IDE5.5 L13, MDC-IV L8, IDE-501 T1, IDE8.17 L6, IDE-503 T10

Water requirement (WR)

The water a crop needs over a given period for normal growth under field conditions, whatever the source. It comprises evapotranspiration (consumptive use), unavoidable application losses and special needs — land preparation, transplanting, leaching and percolation.

$$\text{WR} = (\text{ET or Cu}) + \text{application losses} + \text{special needs}$$

WR is the demand; it is met by irrigation water (IR), effective rainfall (ER) and the soil-profile contribution (S, including a shallow water table):

$$\text{WR} = \text{IR} + \text{ER} + \text{S so field irrigation requirement IR} = \text{WR} - (\text{ER} + \text{S})$$

- **Net water requirement** excludes conveyance losses; **total water requirement** includes them. Express either as a depth, or depth-area units per unit area.
- **Farm irrigation requirement** depends on crop needs, crop area and losses in the farm system (mainly seepage); the **outlet command requirement** adds the conveyance and distribution losses of all farms under the outlet.

Field water balance

An itemised statement of gains, losses and storage changes in a field over a given time: **change in storage = total gains − total losses**.

Gains	Losses	Storage
Precipitation; irrigation; occasionally runoff from higher land; capillary rise from a shallow water table	Surface runoff; deep percolation (drainage); evaporation; transpiration	Change in soil moisture; change in plant water

Net and gross irrigation requirement

- **Net irrigation requirement (NIR)** — water needed to bring the effective root zone back to field capacity: field capacity minus the moisture before irrigation, excluding precipitation, carry-over moisture and groundwater contribution.

$$\text{NIR} = \text{field capacity} - \text{available soil moisture content}$$

$$\text{Layer-wise: } d = \Sigma \text{ (from } i = 1 \text{ to } n) \ [(M_fci - M_bi) \ / \ 100] \times A_i \times D_i$$

d = net depth to apply (cm); M_fci = field-capacity moisture (%) of layer i; M_bi = moisture before irrigation (%); A_i = bulk density of layer i; D_i = depth of layer i (cm); n = number of layers in the root

zone.

> Two layers: 0–30 cm (FC 24%, before 15%, bulk density 1.4) and 30–60 cm (FC 22%, before 16%, 1.5). d = (9/100 × 1.4 × 30) + (6/100 × 1.5 × 30) = 3.78 + 2.70 = 6.48 cm.

- **Gross irrigation requirement (GIR)** — the total to apply so that the NIR enters and stays in the root zone: NIR plus application and other losses.

$$\text{GIR} = \text{NIR} / \text{field efficiency}$$

> NIR 8 cm at 75% field efficiency: GIR = 8 / 0.75 = 10.67 cm (the notes round down to 10.66).

- **Diversion requirement (delta at the head of the main canal)** — the requirement at the source: NIR plus transit losses plus operational wastage; called gross water duty when in similar units.
- **Farm delivery requirement (delta at farm)** — the requirement at the head of the farm = consumptive use + percolation − effective precipitation.

Duty of water

The relation between the area irrigated and the water used to mature the crop — expressed as a depth, depth-area per unit area, area per unit discharge or volume, or volume or discharge per unit area.

Duty	Measured at
Net (farm) duty	The farm
Gross (diversion) duty	The source of diversion
Outlet (lateral) duty	The outlet or head of the water course
Distributary-head duty	The head of a distributary or tertiary canal
Designed duty	Assumed in planning to fix channel capacities

$$\text{Traditional duty–delta relation: } D = 8.6\ B / \Delta$$

D = duty (ha per cumec); B = base period (days) — the whole crop season, ending about 15 days before harvest; Δ = delta (m). (The exact constant is 8.64 = 86,400 s/day ÷ 10,000 m²/ha.) This approach is now replaced by the crop-water-requirement method.

> Wheat with a base period of 120 days and delta 0.40 m: D = 8.64 × 120 / 0.40 ≈ 2592 ha per cumec.

$$\text{Irrigation period (days)} = \text{net soil moisture available between irrigations} / \text{peak-period moisture-use rate}$$

The irrigation period — days allowed for one irrigation over the design area at peak use — sets the system capacity and equipment design.

Irrigation scheduling

- Scheduling decides the number and timing of irrigations, which differ widely with crop and stage. The **critical stage** is the one most sensitive to moisture stress; shortage then cuts yield drastically, so when water is scarce irrigate at the critical stages first.

- **Objectives:** optimum intervals that keep growth, yield and quality while using water efficiently. Make the interval as long as possible without harming growth, yield or quality, allowing for leaching of salts, puddling and soil condition. Soil, plant and climate act singly or together on water use.
- **Water in excess** — aim for maximum yield per unit land: meet full requirements, irrigate intensively, raise cropping intensity, grow higher-water crops and use surface and groundwater together — but beware drainage problems from over-irrigation.
- **Water deficient (arid and semi-arid regions)** — aim for maximum production per unit of water: lower irrigation intensity, cover the most area, irrigate at critical physiological stages, use alternate sources conjunctively — but under-irrigation can upset the salt balance.
- **Table 8.3** of the textbook lists, for common North-Indian crops, the sowing time, number of irrigations, total water requirement (cm), critical stages and timing after sowing.

Plant–water relations and soil moisture

- Water forms 80–90% of plant cells and is a continuous liquid from roots to leaves: root hairs → xylem → mesophyll → sub-stomatal cavities, then out through the stomata as vapour by diffusion. Water also leaves the field by surface evaporation, runoff and deep percolation.
- Water controls cell division and enlargement, photosynthesis and metabolism. Falling water content brings loss of turgor, wilting, stomatal closure, less photosynthesis and, eventually, death. Less than 5% of the water absorbed is used physiologically; most is transpired.
- **Monocots** (wheat, barley, maize, sorghum, millets) have a condensed stem early in growth; **dicots** (peas, gram, mustard, tomato, potato, tapioca) have a distinct stem from the start. Leaves, through their stomata, regulate water-vapour loss and CO_2 exchange.
- **Adequate moisture** promotes root hairs, normal growth and replacement of transpired water. **Excess water** cuts oxygen, keeps the soil cold, invites disease, and standing water can badly damage crops such as maize.

Crop (effective root zone)	Depth
Pasture grasses and vegetables	Up to 45 cm
Vegetable crops	~60 cm
Small grains, maize, seasonal crops	70–120 cm
Lucerne and orchards	2.0–2.5 m or more

Moisture extraction pattern (Fig. 8.13): in deep uniform soils the top 25% of the root zone supplies about 40% of the water used, and each deeper quarter progressively less.

How much and how often

Root zone (Table 8.4)	Sandy (cm)	Loamy (cm)	Clayey (cm)
Shallow (< 60 cm)	2.5–5	5–7.5	7.5–10
Medium (60–90 cm)	5–7.5	10–15	15–20
Deep (1.8–2.4 m)	10–15	20–25	25–35

Time required t = quantity of water for the desired depth / discharge rate

Example 8.3 (border strip): the time works out to 1.19 hours ≈ 1 hour 11 minutes. For instance, a 10 m × 100 m border needing 6 cm (60 m³) from a 14 L/s stream: 60,000 / 14 = 4286 s ≈ 1.19 h.

- Apply enough to refill the root zone to field capacity plus unavoidable losses, as a depth based on the consumptive-use rate and available moisture.
- **Frequency** is set by the available moisture and the ET rate: ensure moisture at planting; irrigate when 50–60% of the available moisture is depleted; young crops use little, flowering and fruiting crops much more; plan economically and in step with the water supply.

Sources: ICAR eCourse 'Irrigation Engineering' (IIT Kharagpur) lessons on soil–water–plant relations, evapotranspiration, irrigation efficiencies, and surface, sprinkler and drip irrigation. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Field irrigation requirement IR equals

(a) WR + ER + S
(b) WR − (ER + S)
(c) ET − WR
(d) WR × efficiency

Q2. With NIR = 8 cm and field efficiency 75%, GIR is about

(a) 6 cm
(b) 8.75 cm
(c) 10.67 cm
(d) 14 cm

Q3. Duty D, base period B (days) and delta Δ (m) are related by

(a) D = 8.64 Δ/B
(b) D = 8.64 B/Δ
(c) D = B Δ/8.64
(d) D = 864 B Δ

Q4. Duty measured at the head of a water course is the

(a) net duty
(b) gross duty
(c) outlet duty
(d) designed duty

Q5. The top quarter of the root zone supplies about ___ of the water used

(a) 10%
(b) 25%
(c) 40%
(d) 70%

Q6. Effective root zone of lucerne and orchards is about

(a) 45 cm
(b) 60 cm
(c) 1 m
(d) 2.0–2.5 m or more

Answers

Q1 — (b) WR − (ER + S)
Q2 — (c) 10.67 cm
Q3 — (b) D = 8.64 B/Δ
Q4 — (c) outlet duty
Q5 — (c) 40%
Q6 — (d) 2.0–2.5 m or more